

JEREMY GINZBURG

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EDUCATION

Duke University (Pratt School of Engineering), Durham, NC

May 2029

BSE Candidate: Biomedical Engineering & Mechanical Engineering | *Minor:* AI & Machine Learning

- **Honors:** 3.9 GPA, Dean's List with Distinction
- **Relevant Coursework:** Quant Physiology & Biostats, Mechanics of Solids, Engineering Design, Innovation & Entrepreneurship

RELEVANT EXPERIENCE

Undergraduate Research Assistant, Vo-Dinh Lab, Duke University

November 2025 - May 2026

- Executed molecular and cell-based workflows in cutting-edge biophotonics and biomedical diagnostic research under the mentorship of 3 PhD candidates
- Mastered and independently operated high-precision instrumentation for PCR, DNA sequencing, and cell culture maintenance, consistently ensuring sterile environments
- Processed 120+ high-throughput biological samples across multiple independent experimental runs, maintaining strict contamination control and high data reproducibility
- Troubleshoot and optimized assay protocols, contributing to data collection pipelines utilized in ongoing biosensing device development

Research Volunteer, Kunda Lab, St. John's University

June 2023 - August 2024 (Summers)

- Synthesized and evaluated 30+ novel lipid-based formulations to optimize drug delivery systems using solid lipid nanoparticles and nanostructured lipid carriers
- Characterized nanocarrier properties via analytical testing (particle size, zeta potential, encapsulation efficiency) to support data-driven optimization of controlled-release profiles
- Authored a comprehensive research manuscript on formulation stabilities and presented at various science symposiums, including Regeneron Science Talent Search

PROJECTS

EMG-Controlled Prosthetic Hand, Independent Project

June 2026 - Present

- Independently designing and prototyping an EMG-controlled prosthetic hand that converts forearm muscle activity into servo-driven finger actuation
- Developing an Arduino-based control system using a MyoWare EMG sensor and establishing an optimized analog threshold algorithm for real-time motor response
- Utilizing MATLAB to visualize raw EMG signal data, identifying noise profiles and mapping out signal-conditioning algorithms to improve future motor stability
- Modeling custom mechanical components in SolidWorks and fabricating assemblies via 3D printing to seamlessly integrate embedded electronics with mechanical joints

Automated Test Tube Opener, Duke University David Lab (Client-Sponsored)

August - December 2025

- Co-engineered an automated, hands-free device tailored for the David Lab at Duke to streamline high-throughput laboratory workflows and mitigate operator injury risks
- Modeled intricate moving assemblies in Onshape, implementing custom spring-driven gripper mechanisms and rack-and-pinion features optimized for robust mechanical advantage
- Validated device performance and efficiency at 90%+ by executing stress testing across 25+ consecutive operational durability cycles without component failure
- Utilized rapid prototyping (3D printing) and compression molding to iteratively refine component fits and fabricate custom silicone fixtures that stabilized test tubes against rotational torque

SKILLS

Software & Programming: Python, MATLAB, C/C++ (Arduino), Onshape, SolidWorks, Git, MS Excel, Powerpoint
Engineering & Prototyping: Embedded Systems, Sensor Integration, Rapid Prototyping (FDM 3D Printing), Silicone Compression Molding, Electromechanical Systems, GD&T

Laboratory and Biological: PCR, DNA Sequencing, Cell Culture (Aseptic Technique), Micropipetting, Biosensing Diagnostics

Machine Learning & Analytics (In Progress): Data Visualization, Signal Processing (EMG Noise Filtering), Statistical Applications